

BENEDICT E. UMEH

Senior Full Stack Engineer | AI Conversational Systems Architect | Customer Experience Technology

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 7+ years delivering production-ready, scalable applications across logistics, e-commerce, hospitality, healthcare, and fintech. I specialise in AI-powered conversational systems that transform customer experience—designing multi-agent chatbots using OpenAI GPT-4o and Google Gemini with function calling, creating interactions that are warm, empathetic, naturally human, and commercially effective.

What sets me apart: deep technical engineering expertise combined with 2+ years frontline hospitality operations. I understand intimately what makes customer interactions feel natural versus robotic, how to handle escalations with empathy, and how to maintain brand voice consistency. Technical foundation spans TypeScript, Python, Java, Node.js, Vue.js, React, and AWS. MSc Data Science enables rigorous, data-driven approach to CSAT improvement.

TECHNICAL EXPERTISE

AI & Conversational Systems: OpenAI GPT-4o, Google Gemini API, Prompt Engineering, Function Calling, Multi-Agent AI Workflows, Conversation Flow Design, AI QA, Tone & Empathy Analysis, Escalation Logic

Languages & Frameworks: TypeScript, JavaScript, Python, Java, PHP | Node.js, Express.js, Vue.js, React 18, Spring Framework, WebSockets, REST APIs

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, CloudWatch), Docker, CI/CD (GitHub Actions, Jenkins), Railway, Microservices

Databases: PostgreSQL, MongoDB, MySQL, Redis, Drizzle ORM, Query Optimization, Data Modeling, ETL Pipelines

Analytics & CX: Power BI, GA4, A/B Testing, CSAT/NPS Measurement, Brand Voice Consistency, Customer Journey Optimization, Agile/Scrum

PROFESSIONAL EXPERIENCE

Founder & Lead AI Engineer

UK Innovator Founder Visa Assistant – AI SaaS Platform | Leeds, UK | November 2024 – Present

- Architected enterprise-grade AI SaaS platform with 109+ production tools across 8 categories (Compliance, Documentation, Team, Business, Financial, Growth, Innovation, Defence), achieving 100% OISC regulatory compliance.
- Engineered Multi-Agent AI Chatbot System with 4 specialized conversational agents (Nova-Innovation, Sterling-Financial, Atlas-Growth, Sage-Compliance)—each with distinct personality and escalation logic. Designed responses to be warm, empathetic, and naturally human.
- Built comprehensive AI quality assurance framework: systematically review interactions for tone, empathy, brand voice consistency. Maintain structured library of best-in-class responses for continuous prompt refinement.
- Implemented data-led approach: identify AI failure points, update prompts, measure impact. Reduced user research time by 70%.
- Developed real-time analytics dashboard tracking sessions, tool usage patterns, activity heatmaps. Built tier-based access control with Stripe integration.

Stack: TypeScript, React 18, Node.js, PostgreSQL, Drizzle ORM, Google Gemini API, Stripe, Railway, WebSockets

Founder & Lead Software Engineer

StackRise Ltd – Enterprise SaaS Platform | Manchester, UK | December 2024 – Present

- Designed STAX AI Chatbot using OpenAI GPT-4o with function calling—handling complex queries with warmth and operational precision. Engineered to never sound robotic.
- Built AI quality assurance framework: reviewed voice/digital interactions for tone, empathy, brand voice. Maintained library of best-in-class responses and failure cases.
- Implemented 3 CRM integration tools through function calling with validation systems to flag incorrect advice or compliance risks.
- Architected full-stack platform with 50+ API endpoints, OAuth 2.0/JWT auth, reducing payment completion time by 90%. Optimized API response times by 60%.

Stack: TypeScript, Python, Node.js, PostgreSQL, OpenAI GPT-4o, Stripe, AWS, Docker, Jest

Guest Service Assistant

Ibis Styles Leeds City Centre Arena | Leeds, UK | September 2023 – Present

- Manage all night shift front-desk operations for 200+ room hotel with consistent warmth, clarity, and empathy. Handle check-ins, complaints, and daily reporting.
- Self-initiated AI Virtual Hotel Concierge automating 200+ guest queries—reduced call volume by 35% while maintaining high satisfaction.
- Developed deep understanding of customer communication patterns—what makes responses feel natural vs robotic, tone perception, effective escalation handling.

Full Stack Developer (Contract)

Qalhata Technology – AI/Autonomous Systems | Cheshire, UK | November 2024 – Present

- Lead development for enterprise AI/autonomous systems clients with end-to-end analytics pipelines and executive dashboards.
- Built automated reporting using Python and Power BI, reducing stakeholder turnaround from 5 days to 1 day (80% improvement).
- Integrated GA4 with custom event tracking. Mentor junior developers on quality standards and Agile methodology.

Web Developer

Deskstones Ltd | Leeds, UK | February 2024 – Present

- Led corporate website rebuild achieving 45% page load improvement through performance optimization and query tuning.
- Increased SEO visibility by 30% and conversion rate by 25% through A/B testing and UX optimization.

Freelance Software Engineer & Founder

bhenmedia.com | Leeds, UK & Lagos, Nigeria | 2017 – Present

- Founded web development consultancy delivering 50+ applications for e-commerce, hospitality, healthcare, and logistics clients.
- Built e-commerce platforms with Stripe/PayPal payment processing, order tracking, inventory management.
- Migrated legacy PHP applications to modern TypeScript/Node.js architectures. Notable: Ebluniversal.com, Slot.ng, Vivano.com.ng

AI & CONVERSATIONAL SYSTEMS PROJECTS

Multi-Agent AI Visa Assistant System | November 2024 – Present

- Engineered 4 specialized AI agents with distinct personalities, expertise domains, and conversation flows

- Built agentic AI using function calling with multi-step tool orchestration across 6-stage analysis workflows
- Implemented validation systems ensuring API actions complete correctly; built compliance risk flagging

STAX PhD-Level AI Assistant | December 2024

- Designed enterprise AI chatbot for natural, human-like flows—never robotic, maintaining warmth and conversational naturalness
- Built conversation quality framework with systematic review for tone, empathy, accuracy. Edge case handling and escalation logic

EDUCATION

MSc Data Science | Leeds Beckett University, UK | 2023

BSc Information Technology & Business Information Systems | Middlesex University, London | 2017

Advanced Diploma in Software Engineering | Aptech Computer Institute, Lagos | 2016

CERTIFICATIONS

IBM Data Analyst Professional | Google Data Analytics Professional | Google Analytics IQ | AWS Cloud Practitioner (In Progress)

KEY ACHIEVEMENTS

- Built production multi-agent AI chatbot with 4 specialized assistants—natural, empathetic responses with proactive engagement
- Engineered 109+ AI-powered tools achieving 100% OISC compliance across 8 business categories
- Reduced hotel call volume by 35% through AI virtual concierge with quality conversation design
- Optimized API response times by 60% through data-driven analysis and intelligent caching
- Delivered 50+ applications across e-commerce, logistics, healthcare with consistent quality